Novel Application of Stereotaxically Guided Dental Germinative Photodynamic Regulation in *Chinchilla lanigera* with Refractory Malocclusion.

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7th January 2026

Abstract

Dental Germinative Photodynamic Regulation (dGPR) is a theoretical photodynamic technique for the treatment of refractory malocclusion in hypselodont species, primarily rodents and lagomorphs, that results in a reduced rate of eruption. This proposal is focused on chinchillas (Chinchilla lanigera) due to their high incidence rate of malocclusion (Crossley, 2001). This is an important development as current techniques for refractory malocclusion are palliative, with - according to statistical projections - practically no patients living for their full lifespan. Extrapolation from current data on anesthetic risk suggests that a skilled veterinary surgeon could use this procedure to achieve a significantly improved survival rate (Brodbelt et al., 2008).

Background and Clinical Need

Chinchillas (*Chinchilla lanigera*) possess hypselodont dentition, characterized by continuous eruption throughout life (Hisey et al., 2024). In chinchillas, the germinative tissues of the teeth are located at the base of the reserve crown (Brenner et al., 2005), the position of which varies by tooth. The reserve crowns of the mandibular teeth are located deep in the mandible, extending to the ventral margin of the mandible; the reserve crowns of the maxillary molars are located just below the alveolar floor, bordering the infraorbital region and orbital cavity; the reserve crowns of the maxillary incisors are located just below the surface at the lateral sides of the incisive bone (Natural History Museum of Los Angeles County, 2020). Dental malocclusion, where the rate of eruption exceeds the rate of wear, is a pervasive and often fatal condition in this species (Mans & Donnelly, 2021).

The current standard of care involves repeated coronal reduction under general anesthesia (Mans & Donnelly, 2021). Because coronal reduction only addresses the clinical signs of malocclusion rather than the underlying pathology, treatments must be repeated at varying

https://doi.org/10.5281/zenodo.18203469

intervals for the remainder of the animal's life in order to control pain and allow them to eat unaided (Mans & Donnelly, 2021). This approach is strictly palliative and carries significant cumulative anesthetic risk, as small mammals have higher perianesthetic mortality rates compared to dogs and cats (Brodbelt et al., 2008). The common alternate approach, surgical extraction, is invasive, physically traumatic, and prone to severe complications (Lennox et al., 2021). To be specific, a chinchilla diagnosed at age four and managed via coronal reduction theoretically faces more than 90 general anesthetic events within its natural lifespan (Legendre, 2002). Brodbelt et al. (2008) reported a mean per-anesthetic mortality rate of 3.29% in chinchillas (n=334). At this rate, the cumulative statistical probability of anesthetic death over a lifetime of palliative care exceeds 95%. While clinical reality is often truncated by earlier euthanasia or fatal secondary effects (Crossley, 2001), this trajectory demonstrates that the current standard of care is statistically terminal. Thus, a curative intervention that modulates the rate of tooth growth is critically needed.

Proposed Solution and Hypothesis

I propose Dental Germinative Photodynamic Regulation (dGPR), my adaptation of stereotaxically guided interstitial photodynamic therapy specific to hypselodont dentition, to achieve precise partial ablation of the apical germinative tissue.

Hypothesis: dGPR can safely and effectively reduce the volume of active odontogenic germ cells, thereby slowing the rate of tooth eruption to restore the equilibrium between growth and wear, reducing or eliminating the need for repeated anesthetic interventions.

Unlike existing ablative techniques, this method aims to preserve functional tooth structure while restoring physiological equilibrium.

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Rationale for dGPR

Initial theoretical concepts involving direct thermal laser ablation are clinically non-viable due to a high risk of collateral thermal damage to osseous tissue (Eriksson & Albrektsson, 1983), which is likely exacerbated due to the sub-millimeter thickness of several areas of the bone, including the maxillary alveolus (Natural History Museum of Los Angeles County, 2020).

Photodynamic Therapy (PDT) utilizes a photosensitizing agent activated by non-thermal light, inducing cell death via reactive oxygen species (Agostinis et al., 2011). This non-thermal mechanism allows for intraosseous treatments without collateral osteonecrosis from heat damage (Bisland & Burch, 2006). The dual-selective nature of this procedure enhances the precision of the ablation application. The biological selectivity of photosensitizers, which preferentially accumulate in metabolically active tissues, would cause them to gather in the germinative stem cells due to their high rate of division (Dolmans et al., 2003; Harada et al., 1999). The spatial selectivity of the light delivery via interstitial optic fiber would allow for tight control of the ablation area within the biologically selected tissue. Additionally, the procedure would be as minimally invasive as currently possible, as light delivery can be achieved via small-gauge interstitial probes (Harris et al., 2017; Sahovaler et al., 2024).

A synergistic benefit of photodynamic therapy is that it can modulate the microvascular environment within the targeted area (Krammer, 2001; Suzuki et al., 2020). This vascular regulation restricts the metabolic supply to the niche, theoretically acting as a metabolic limiter and preventing the compensatory proliferation of surviving stem cells. This reduction in proliferation would then reduce the potential eruption rate governed by the reduced oxygen supply, and the non-necrotic resolution allows for titratable, iterative treatments if required.

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A key objective for further study is to optimize dosimetry, potentially through fractionated light delivery, to preferentially induce apoptosis while minimizing necrosis. This reduces the risk of severe inflammation and promotes controlled tissue reduction.

Methodology Outline: Stereotaxic Guidance

The deep intraosseous location of the germinative tissue necessitates sub-millimeter accuracy. Small mammals like chinchillas are particularly sensitive to anesthetic use, thus the guide designed for this protocol holds the head upright, allowing two surgeons to work in tandem for significantly reduced anesthetic time. My proposed workflow integrates advanced imaging alongside a patient-specific Auriel Stereotaxic Guide (for safety and design information, refer to the separate document: Patient Specific Stereotaxic Guide Specifications for Dental Germinative Photodynamic Regulation):

1. Computed Tomography (CT) will be used to map the 3D volume and location of the target germinative tissue, given its proven efficacy over other radiological methods for rodents (Capello, 2016).

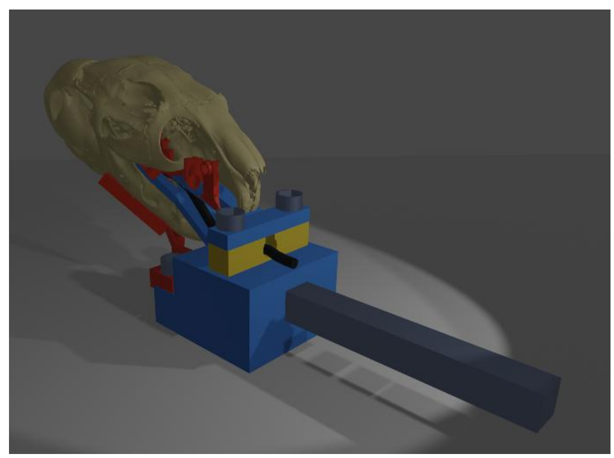

Figure 1: Auriel Stereotactic Guide, Alpha Prototype Render

2. A 3D reconstruction and dosimetry plan will be developed. Fabrication of patient-specific, 3D-printed stereotaxic guides are necessary to ensure the precision of the trajectory of the interstitial probe (Shinn et al., 2021).
3. The patient will be administered with a photosensitizer, with time differing dependent on the selected photosensitizer. Chinchillas are likely more resistant to

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the associated photosensitivity effects due to their thick coat (Wilcox, 1950), but they should be kept in a low light environment to protect their eyes, ears, nose, and paws. Being in a low light environment is not uncommon in their daily lives, as they typically use hides, thus there should be little stress associated with this.

4. This is then followed by the surgical procedure:
    - The patient is intubated, and their pharynx is packed with sterile gauze to prevent aspiration.
    - This is followed by standard aseptic preparation of the operative site, as it is shaved for clean contact with external guides.
    - The stereotaxic guide is then fitted, first the intra-oral mandibular block is placed. The maxillary block is then fixed to the maxilla. The intubation tube is placed into the intubation block which is inserted between the mandibular and maxillary blocks. Those blocks are then affixed to one another via M3 screws. Finally, the external mandibular blocks are attached to the intra-oral mandibular block.
    - The guides are then used to make incisions and drill in the predetermined trajectories towards the germinative apices, these guides are lined with surgical steel, for structural integrity given drill proximity and to isolate the operational zone optically. A high torque/low speed drill alongside cooled saline will be used to manage temperature and prevent osteonecrosis caused by heat, which can occur at temperatures as low as 47°C (Eriksson & Albrektsson, 1983).

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- The optic fiber (likely a capped cylindrical diffuser for balanced coverage with a dark zone for avoidance of sensitive tissues) is inserted into the germinative apex, and the photodynamic diode activated. Fractionated light delivery will be ideal to mitigate heat and maximize reactive oxygen species efficiency.
- The previous two steps can be repeated for every tooth, the guide makes it possible for two veterinarians to work in tandem, one managing each side of the face, to halve the time spent under anesthesia.
- Once a germinative apex has been given the calculated light dosage, the optic fiber is removed, the drill burr in the bone is packed with resorbable gelatin sponge, and associated incisions are sutured. Skin sutures should be subcuticular to prevent damage via scratching.
- Standard analgesia, such as Meloxicam (Sadar & Mans, 2023), should be administered immediately post-operatively, as chinchillas are sensitive to pain and may refuse to eat, causing gastrointestinal stasis, which can be fatal.
- Prophylactic antibiotics approved for use in chinchillas, an example being Enrofloxacin (Wang et al., 2025), should also be administered post-operatively to prevent infection.
- Critical care (Oxbow, Herbivore variant), or another clinically validated syringe food, should be administered via syringe within two hours of waking if the patient does not eat on their own. This rapid intervention is

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necessary for prevention of gastrointestinal stasis and hepatic lipidosis (Ritzman, 2014).

Photosensitizer Selection

For live cases, the photosensitizer delivery method that appears most clinically viable are porphysomes. This is due to their accumulation in inflamed tissues which mirror the enhanced permeability and retention seen in tumors (Philp et al., 2019), which would theoretically concentrate the photodynamic effect on the inflamed, heavily vascularized, germinative niche. Additionally, the accumulation of porphysomes in inflamed tissue may have a rebalancing effect on the tissue, as the most inflamed tissues and most vascularized tissues would be preferentially targeted, which I hypothesize would add a biological targeting mechanism for the most active germinative tissue. Additionally, porphysomes absorb 671nm light most effectively (Lovell et al., 2011), which is scattered by cortical bone and dentin (Shanshool et al., 2022), allowing the photons to pass over target cells multiple times, and acting as a safety feature which prevents damage to cells external to the germinative apex. An additional feature of porphysomes is their fluorescence profile. They begin as nanostructures (vesicles), with little fluorescence (Lovell et al., 2011). When there is uptake of porphysomes in cells, they are dissociated into their active form, which is highly fluorescent. This allows for fluorescence imaging prior to an operation to ensure the germinative apex internalized porphysomes prior to surgery, to ensure no unnecessary operation takes place if the drug delivery fails.

For the cadaver study, this would not be feasible as the blood cannot carry the porphysomes to their destination in the absence of a pulse, thus I have selected methylene blue for its similar light absorbance (~665nm), which would be injected interstitially into the

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germinative apex through the drill path. This would allow assessment of the light dispersion in the space, thus enabling review of the effectiveness of different trajectories.

Risk Mitigation and Contingency

The primary technical risk is the precise calibration of the ablation volume. If ablation is excessive and wear exceeds the slowed eruption rate, the ideal contingency is supplemental or full syringe feeding. This outcome is preferable to the cumulative risks of lifelong anesthesia or euthanasia, which are currently unavoidable in such cases. Due to the non-necrotic resolution of PDT, the structural integrity of the alveolus is predicted to be preserved based on preliminary Monte Carlo simulations (Appendix). This then allows for iteration and titration of the procedure, which enables the clinical team to adjust the eruption rate over multiple sessions, without the trauma or pain of extraction or burring. While iterative therapy does reintroduce additional anesthetic episodes, this finite intervention protocol represents a statistically drastic reduction in cumulative risk compared to indefinite, high-frequency anesthetic required for current palliative management. Limiting the protocol to a series of three sessions, with the necessary three pre-operative diagnostic procedures which would be necessary for CT, confines the cumulative anesthetic risk to <20%, which is an >75% improvement in projected mortality rates when compared to standard palliative care (figure 2),

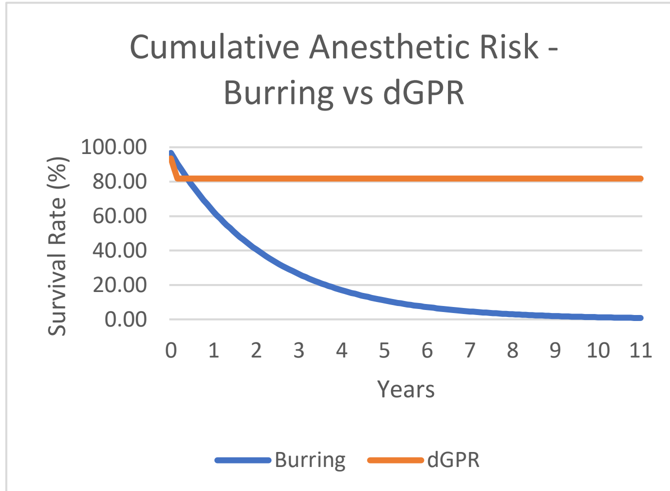

Figure 2: Comparative anesthetic risk assuming 1 burring every 4 weeks (Legendre, 2002). Extrapolated from 3.29% mean anesthetic survival rate in Brodbelt et al. (2008), compared to the anesthetic risk of 3 titrated dGPR procedures, alongside 3 anesthetized preoperative diagnostic procedures, as proposed.

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while simultaneously eliminating the chronic welfare impact of indefinite monthly anesthetic induction.

Research Objectives and Next Steps

The immediate goal is to transition this theoretical framework into a pilot study. The proposed initial phases are:

1. **Phase 1 (Ex Vivo)**: Test the accuracy of 3D-printed surgical guides, and establish preliminary dosimetry parameters (exact light and photosensitizer dosages, and tissue response to the methodology)
2. **Phase 2 (Pilot In Vivo)**: Pending ethical approval, the full dGPR protocol would be performed on clinical cases with guarded or poor prognoses (e.g. candidates for euthanasia) directly before administering humane euthanasia. Both procedures would be performed under the same anesthetic episode, without the patient being awoken.
3. **Phase 3 (In Vivo Survival Studies)**: Assuming photosensitizer and photodynamic efficacy in the pilot study, the treatment would then be performed on advanced in vivo cases with post-operative progress updates.

This research is a translational effort to adapt established biophotonic oncological technology to solve a persistent, critical challenge in exotic companion mammal dentistry through precision, patient specific engineering, which, if successful, could be translated across species with similar dentition.

I am actively seeking academic mentorship and institutional collaboration to progress this work from a theoretical framework to ethical pilot testing. I would be deeply grateful for any

https://doi.org/10.5281/zenodo.18203469

guidance or introduction to faculty with interest in translational oncology or exotic mammal dentistry.

Acknowledgement

I would like to thank Shawn R. S. Del Carmen for the initial conceptual dialogue in which he proposed thermal laser ablation as a potential treatment for refractory malocclusion. These discussions served as the catalyst for my research into other non-thermal, light-based, oncological techniques, which led to my development of this protocol as a safer alternative for germinative tissue modulation.

https://doi.org/10.5281/zenodo.18203469

Appendix: Preliminary Monte Carlo Simulations

Monte Carlo simulations were run to determine the safety profile of differing fluence ranges. These simulations used the console version of MCX (Fang & Boas, 2009) using 25µm voxels and 500 million photons, generated with the use of a workstation using an Intel 13700kf central processing unit, 64GB of RAM, and a Nvidia rtx 4090 graphics card. JSON simulation configuration files, model files, and the programs used to generate the shown images are in an attached file.

The following are Monte Carlo simulations run on areas considered to be of the highest risk to sensitive structures nearby.

Both models used hybrid optical data from various sources. Notably, both models use parameters from rabbit bone, rabbit neural tissue (Shanshool et al., 2022), and human dental pulp data (Fu & Jacques, 2011), as these are the most relevant figures to the simulation. Materials intended to be opaque, like the fiber coating and resin guide with steel inlay, were assumed to be opaque, while the fiber tip was assumed to be near transparent.

The threshold figure for partial apoptosis has a basis in Kato et al. (2017), and the near-complete apoptosis threshold was adapted from Kato et al. (2017) alongside Sahovaler et al. (2024)

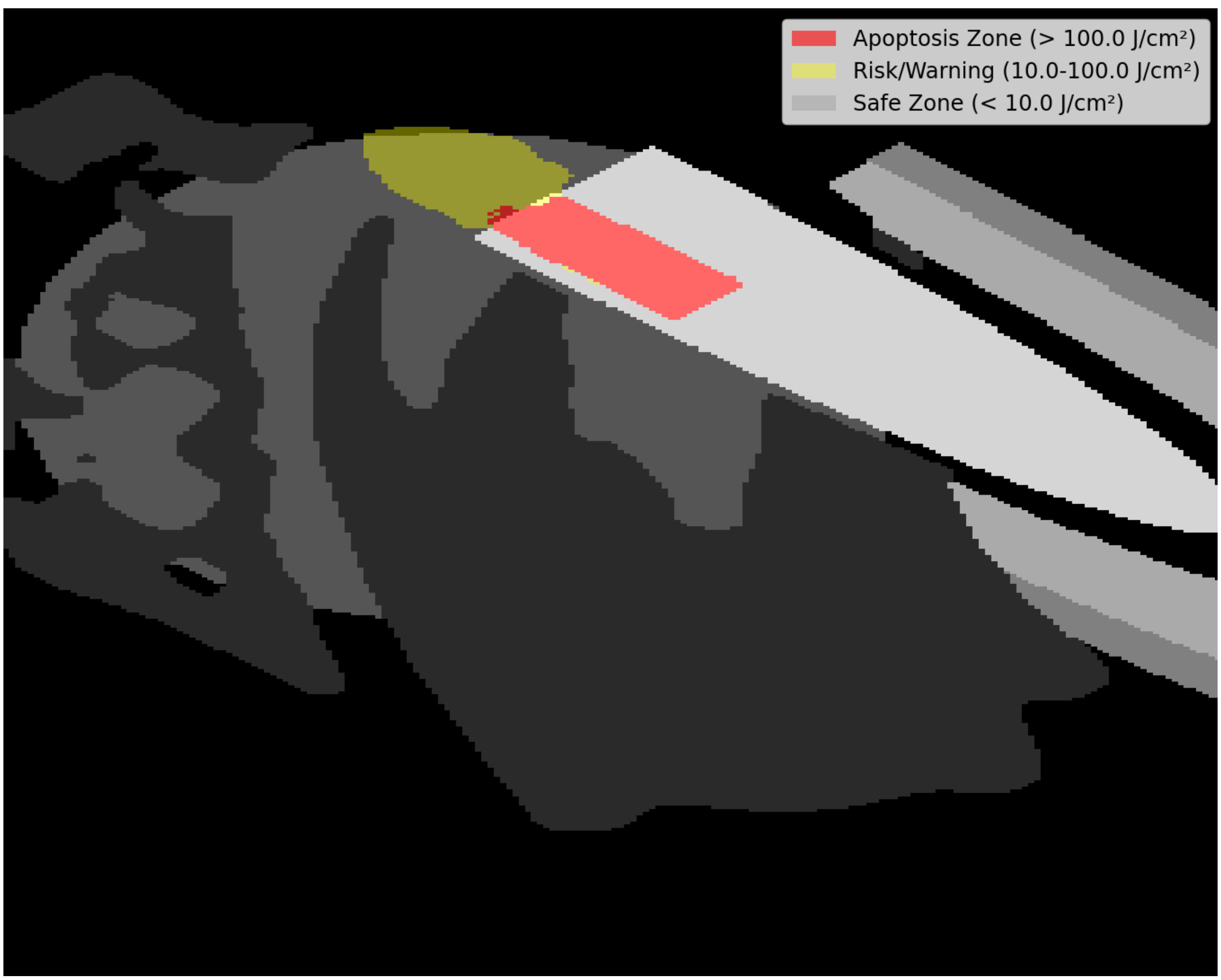

Figure 3: Monte Carlo Simulation of the Posterior Maxillary Molar, using an incident dose of 20J/cm^2

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The posterior maxillary molar was selected due to the thickness of the alveolar bone in the area, as well as its proximity to the cranial vault. Prior simulations were used as design challenges, primarily the reserve crown of the molar, which was penetrated by the drill due to necessity, would have been vulnerable to damage from the diode. This was mitigated via the coating of the sides of the cylindrical diffuser, with a semicircular window left open at the tip of the diode to more precisely target the germinative tissue and supporting vasculature with minimal collateral damage.

Figure 4: Monte Carlo Simulation on the Maxillary Incisor, using an incident dose of 5J/cm^2

The maxillary incisor was selected for its proximity to the nasolacrimal duct. As can be seen, the fluence reached the surrounding bone, however, this is acceptable because of the physiology of bone tissue. Even under inflammatory conditions, osteocytes only intake solutes from the lacunar-canalicular system - and thus only have an intake of particles up to 17nm in size (Wang, 2018), meaning they will never be at risk of saturation with porphysomes due to their approximately 100nm size (Lovell et al., 2011). This acts as a protective effect for the bone, limiting the damage it could take to thermal sources only, which will not be relevant below 50J/cm^2 with fractionated delivery.

Of course, all fluence statistics displayed here are preliminary and will almost certainly be refined with ex vivo and pilot studies. Additionally, the fiber coating can, and likely will, be adjusted to accommodate any vital structures that must be avoided.

https://doi.org/10.5281/zenodo.18203469

It should be noted that the simulations used time intervals as small as 5 nanoseconds, thus these simulations show the intended final light dosage for dentition like this, but not the time interval, as a 5 nanosecond pulse would make an extremely high peak power density, potentially causing a shockwave as the photosensitizer in the pulp absorbs light so quickly, creating pockets of high energy which quickly turn into photothermal and photomechanical action (Anderson & Parrish, 1983), which represent energy not used for photochemical release of reactive oxygen species.

For ideal dosage, the timing would need to be stretched to a much longer interval, like 10ms pulses with 50ms thermal relaxation periods (more detailed calculations would have to be performed to determine the true ideal intervals, preferably with verified optical properties for relevant tissues), which would be barely noticeable to the surgeon and not significantly increase operative time.

This kind of interval schedule could be easily managed by an inexpensive microcontroller circuit, which would activate or inactivate the diode circuit at the precise intervals. By using this microcontroller architecture, the procedure could be further tailored to the patient's specific pathology by changing the overall duration of light delivery while maintaining low material cost. With a procedure like this, the anesthetic time is determined by how fast the surgeon can drill, as each tooth treatment would be 0.6 seconds (or similarly short depending on the patient's pathology), thus reducing patient risk by truncating procedure time.

https://doi.org/10.5281/zenodo.18203469

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